

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Pattern Recognition and Text Processing

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Initiation (L1)	Assessment:	Assessment Tool 1 – Pattern Recognition and Text Processing
Weightage:	40% of CIA		
CO5 Statement:	Analyse regular expression patterns. (BL-3)		
Student Name:	P. Suvayana	Reg. No.:	192424303
Section / Batch:	AI & DS	Date:	14-07-2026

Code of Conduct

P. Suvayana (192424303) Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. Suvayana

Instructions

- Show all intermediate steps, calculations, state transitions, and reasoning wherever applicable.
- Use only the Python re (Regular Expressions) module to solve the given problems.
- Clearly mention the Regular Expression (Regex) pattern used before writing the corresponding Python program.
- Display the required output for each question, including extracted values, validation results, counts, and positions wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Regular Expression Pattern Generation	Pattern Matching Information Extraction	1
Search for a String	Keyword Search and Text Analysis	2
Split the documentation into sentences and words	Text Tokenization and Sentence Segmentation	3
Email and Strong Password Validation	Data Validation and Format Verification	4
Compile Once, Validate Many	Pattern Reusability and Performance Optimization	5

Question 1:

For the given student details formulate Regex patterns for the following:

```
import re
text = """
```

Student Details:

Name: John Doe

Email: john.doe123@gmail.com

Mobile: +91-9876543210

Password: P@ssw0rd123

Date of Birth: 15/08/2004

Register Number: 23AIML1056

Department: Artificial Intelligence and Machine Learning

```
"""
```

Email

```
email = re.findall(r'[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}', text)
```

Mobile Number

```
mobile = re.findall(r'\+91-\d{10}', text)
```

Strong Password

```
password = re.findall(r'(?=[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@$%^&*!]).{8,}', text)
```

Date of Birth

```
dob = re.findall(r'\d{2}/\d{2}/\d{4}', text)
```

Register Number

```
reg = re.findall(r'\b\d{2}[A-Z]{4}\d{4}\b', text)
```

```
print("Email:", email)
```

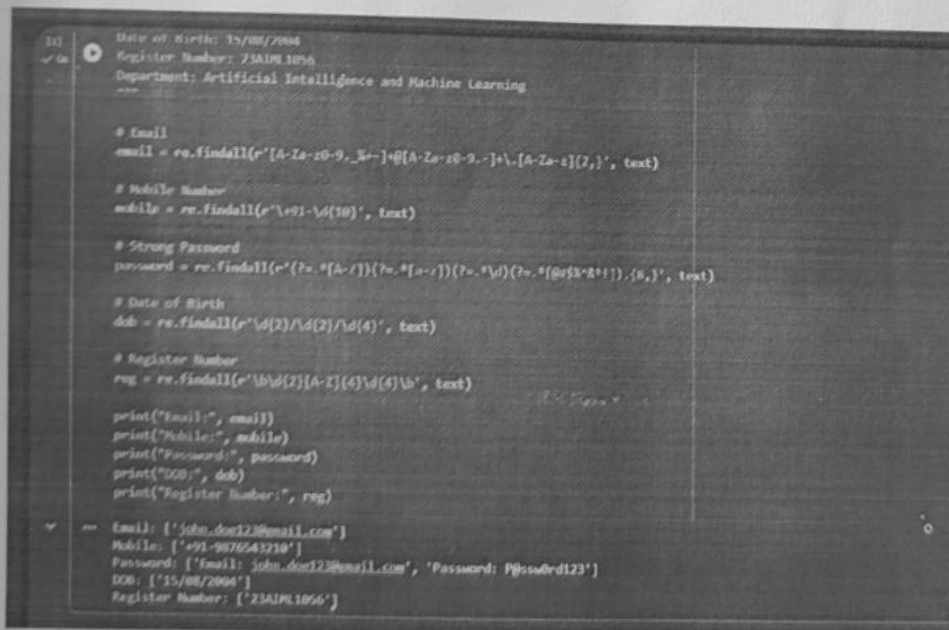
```
print("Mobile:", mobile)
```

```
print("Password:", password)
```

```
print("DOB:", dob)
```

```
print("Register Number:", reg)
```

OUTPUT:



```
131 Date of Birth: 15/08/2004
132 Register Number: 23AIML1056
133 Department: Artificial Intelligence and Machine Learning
134 """

135 # Email
136 email = re.findall(r'[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}', text)

137 # Mobile Number
138 mobile = re.findall(r'\+91-\d{10}', text)

139 # Strong Password
140 password = re.findall(r'(?=[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@$%^&*!]).{8,}', text)

141 # Date of Birth
142 dob = re.findall(r'\d{2}/\d{2}/\d{4}', text)

143 # Register Number
144 reg = re.findall(r'\b\d{2}[A-Z]{4}\d{4}\b', text)

145 print("Email:", email)
146 print("Mobile:", mobile)
147 print("Password:", password)
148 print("DOB:", dob)
149 print("Register Number:", reg)

150 Email: ['john.doe123@gmail.com']
151 Mobile: ['+91-9876543210']
152 Password: ['P@ssw0rd123']
153 DOB: ['15/08/2004']
154 Register Number: ['23AIML1056']
```

Question 2:

Coding:

```
text = """Artificial Intelligence (AI) is transforming industries across the world.  
AI is used in healthcare to assist doctors in diagnosis,  
banking to detect fraud, and in education to provide personalized learning experiences.  
Many companies invest heavily in AI research because AI improves efficiency.  
As AI continues to evolve, professionals with AI skills are in high demand."""  
word = "AI"
```

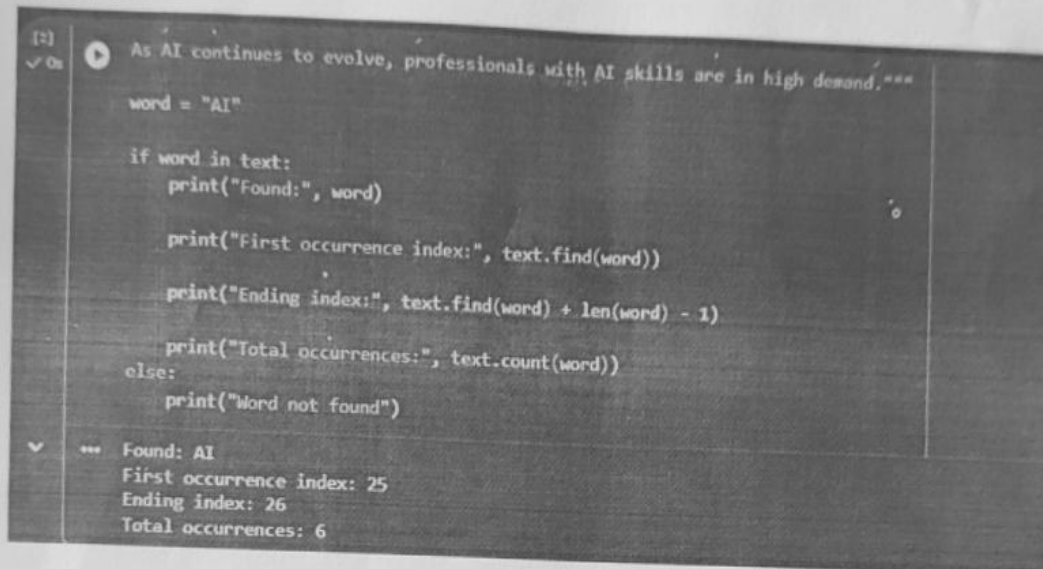
if word in text:

```
    print("Found:", word)  
    print("First occurrence index:", text.find(word))  
    print("Ending index:", text.find(word) + len(word) - 1)  
    print("Total occurrences:", text.count(word))
```

else:

```
    print("Word not found")
```

OUTPUT:



```
[2] ✓ On As AI continues to evolve, professionals with AI skills are in high demand.  
word = "AI"  
  
if word in text:  
    print("Found:", word)  
  
    print("First occurrence index:", text.find(word))  
  
    print("Ending index:", text.find(word) + len(word) - 1)  
  
    print("Total occurrences:", text.count(word))  
else:  
    print("Word not found")  
  
✓ *** Found: AI  
First occurrence index: 25  
Ending index: 26  
Total occurrences: 6
```

Question 3:

import re

```
text = """Artificial Intelligence (AI) is transforming industries across the world.  
AI is used in healthcare to assist doctors in diagnosis,  
banking to detect fraud, and in education to provide personalized learning experiences.  
Many companies invest heavily in AI research."""
```

Split into sentences

```
sentences = re.split(r'[.!?]', text)
```

```
print("Sentences:")
```

```
for s in sentences:
```

```
    if s.strip():
```

```
        print(s.strip())
```

Split into words

```
words = re.split(r'\s+', text)
```

```
print("\nTotal Words:", len(words))
```

```
print("\nWords:")
```

```
print(words)
```

OUTPUT:

```
import re

text = """Artificial Intelligence (AI) is transforming industries across the world.
AI is used in healthcare to assist doctors in diagnosis,
banking to detect fraud, and in education to provide personalized learning experiences.
Many companies invest heavily in AI research."""

# Split into sentences
sentences = re.split(r'[.!?]', text)

print("Sentences:")
for s in sentences:
    if s.strip():
        print(s.strip())

# Split into words
words = re.split(r'\s+', text)

print("\nTotal Words:", len(words))

print("\nWords:")
print(words)

--- Sentences:
Artificial Intelligence (AI) is transforming industries across the world
AI is used in healthcare to assist doctors in diagnosis,
banking to detect fraud, and in education to provide personalized learning experiences
Many companies invest heavily in AI research

Total Words: 38

Words:
['Artificial', 'Intelligence', '(AI)', 'is', 'transforming', 'industries', 'across', 'the', 'world.', 'AI', 'is', 'used', 'in', 'healthcare', 'to', 'assist', 'doctors', 'in', 'diagnosis,', 'banking', 'to', 'detect', 'fraud,', 'and', 'in', 'education', 'to', 'provide', 'personalized', 'learning', 'experiences.', 'Many', 'companies', 'invest', 'heavily', 'in', 'AI', 'research']
```

Question 4:

```
import re
```

```
email = input("Enter Email: ")
```

```
password = input("Enter Password: ")
```

```
email_pattern = r'^[A-Za-z0-9]+[A-Za-z0-9._%+-]*@[A-Za-z0-9.-]+\.(com|org|edu|in)$'
```

```
password_pattern = r'^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#%$%^&*!]).{8,}$'
```

```
if re.match(email_pattern, email):
```

```
    print("Valid Email")
```

```
else:
```

```
    print("Invalid Email")
```

```
if re.match(password_pattern, password):
```

```
    print("Strong Password")
```

```
else:
```

```
    print("Weak Password")
```

OUTPUT:

```
[4]
✓ 4/6 import re

email = input("Enter Email: ")
password = input("Enter Password: ")

email_pattern = r'^[A-Za-z0-9]+[A-Za-z0-9._%+-]*@[A-Za-z0-9.-]+\.(com|org|edu|in)$'

password_pattern = r'^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#%$%^&*!]).{8,}$'

if re.match(email_pattern, email):
    print("Valid Email")
else:
    print("Invalid Email")

if re.match(password_pattern, password):
    print("Strong Password")
else:
    print("Weak Password")

--- Enter Email: sunayma456@gmail.com
Enter Password: p#suna234$
Valid Email
Weak Password
```

Question 5:

```
import re
```

```
emails = [
```

```
    "john@gmail.com",
```

```
    "abc@yahoo.in",
```

```
    "student.edu@gmail.com",
```

```
    "wrong@email",
```

```
    "123@test.org"
```

```
]
```

```
pattern = re.compile(r'^[A-Za-z0-9]+[A-Za-z0-9._%+-]*@[A-Za-z0-9.-]+\.(com|org|edu|in)$')
```

```
for email in emails:
```

```
    if pattern.match(email):
```

```
        print(email, "-> Valid")
```

```
    else:
```

```
        print(email, "-> Invalid")
```

OUTPUT:

```
import re

emails = [
    "john@gmail.com",
    "abc@yahoo.in",
    "student.edu@gmail.com",
    "wrong@email",
    "123@test.org"
]

pattern = re.compile(r'^[A-Za-z0-9]+[A-Za-z0-9._%+-]*@[A-Za-z0-9.-]+\.(com|org|edu|in)$')

for email in emails:
    if pattern.match(email):
        print(email, "-> Valid")
    else:
        print(email, "-> Invalid")

-- john@gmail.com -> Valid
   abc@yahoo.in -> Valid
   student.edu@gmail.com -> Valid
   wrong@email -> Invalid
   123@test.org -> Valid
```


Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2-3 Marks)	Level 1 – Beginning (0-1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q. No. / Criteria	Conceptual Understanding	Accuracy of Answer	Application of Knowledge	Completeness of Response	Clarity & Presentation	Sub. Total	Total %
1	4	4	4	4	4	20	94.
2	4	4	4	4	4	20	
3	4	4	4	3	3	20	
4	4	4	4	3	3	18	
5	4	4	4	3	3	18	

Faculty In-charge	Course Coordinator	Program Director
<i>[Signature]</i> Signature: _____ Date: 23/7	Signature: _____ Date: _____	Signature: _____ Date: _____